```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

df = pd.read_csv('/content/shop-sales-data.csv')

print("Dataset Loaded Successfully!")
print(df.head())
```

```
Dataset Loaded Successfully!
   year  month shop_id   product_id  sales_qty
0     2      2  shop_0  product_268       2.00
1     2      4  shop_0  product_268       2.00
2     2      9  shop_0  product_268       2.00
3     2      3  shop_0  product_269       0.95
4     2      7  shop_0  product_269       0.95
```

```
print("\nDataset Info")
print(df.info())

print("\nMissing Values")
print(df.isnull().sum())
```

```

Dataset Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 274381 entries, 0 to 274380
Data columns (total 5 columns):
 #   Column      Non-Null Count   Dtype
---  ------      --------------   -----
 0   year        274381 non-null  int64
 1   month       274381 non-null  int64
 2   shop_id     274380 non-null  object
 3   product_id  274380 non-null  object
 4   sales_qty   274380 non-null  float64
dtypes: float64(1), int64(2), object(2)
memory usage: 10.5+ MB
None

Missing Values
year          0
month         0
shop_id       1
product_id    1
sales_qty     1
dtype: int64
```

```python
df = df.dropna()

X = df[['sales_qty']]
y = df['sales_qty']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

model = LinearRegression()

model.fit(X_train, y_train)

print("\nModel Training Completed!")
```

```
Model Training Completed!
```

```python
y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)

print("\nModel Evaluation")
print("----------------------")

print("MAE :", mae)

print("MSE :", mse)

print("RMSE:", rmse)
print("R2 Score:", r2)
```

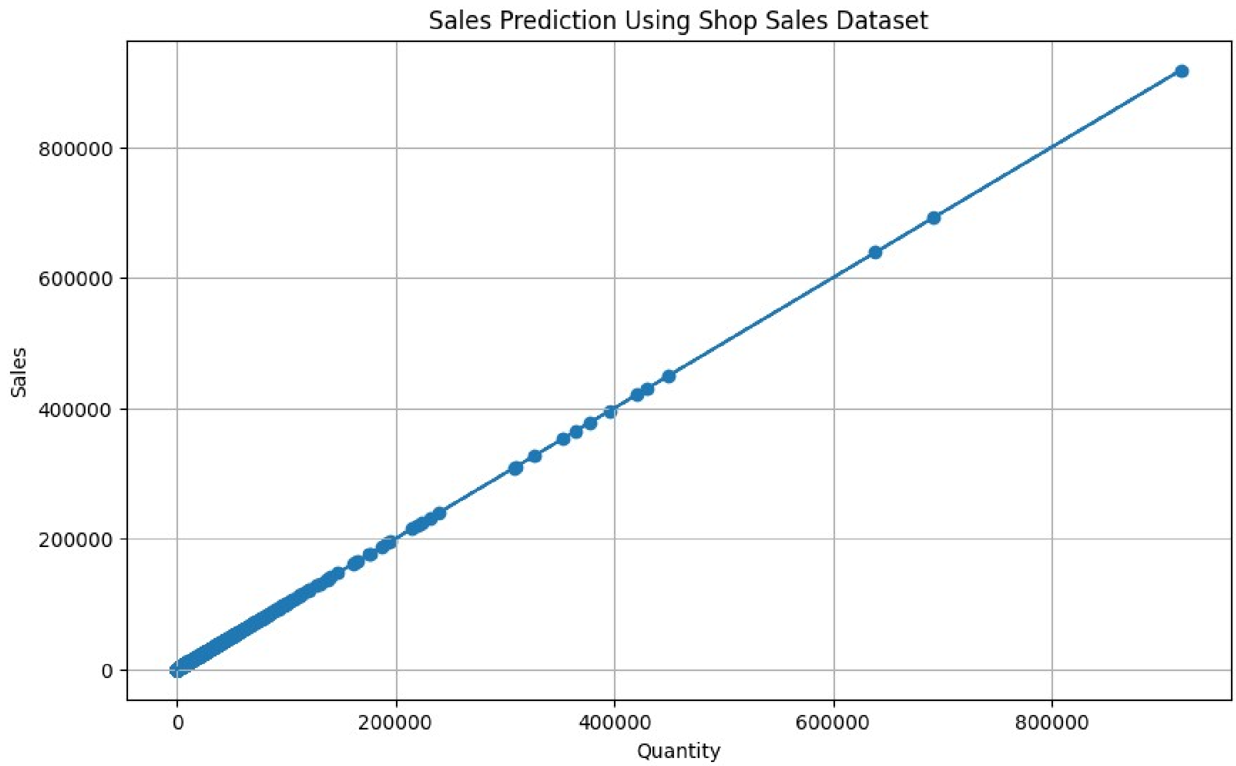
```
Model Evaluation
----------------------
MAE : 8.969478989184528e-14
MSE : 4.0635700753666047e-25
RMSE: 6.374613772901543e-13
R2 Score: 1.0
```

```python
future_quantity = pd.DataFrame({
    'sales_qty': [10, 15, 20, 25]
})

future_sales = model.predict(future_quantity)

print("\nFuture Sales Predictions")
```

```
print("---------------------------")

for qty, sale in zip(future_quantity['sales_qty'], future_sales):
    print(f"Quantity {qty} --> Predicted Sales = {sale:.2f}")
```

```
Future Sales Predictions
---------------------------
Quantity 10 --> Predicted Sales = 10.00
Quantity 15 --> Predicted Sales = 15.00
Quantity 20 --> Predicted Sales = 20.00
Quantity 25 --> Predicted Sales = 25.00
```

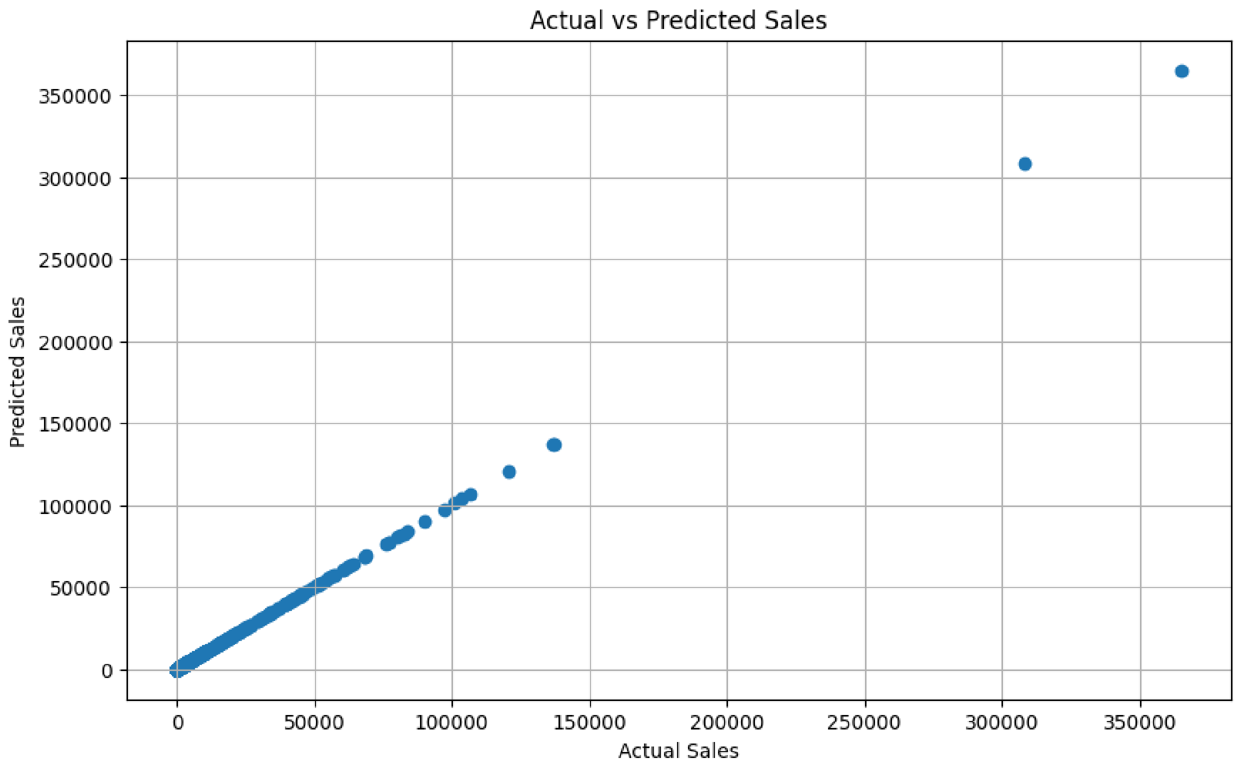
```
plt.figure(figsize=(10,6))

# Actual Data
plt.scatter(X, y)

# Regression Line
plt.plot(X, model.predict(X))

plt.title("Sales Prediction Using Shop Sales Dataset")

plt.xlabel("Quantity")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



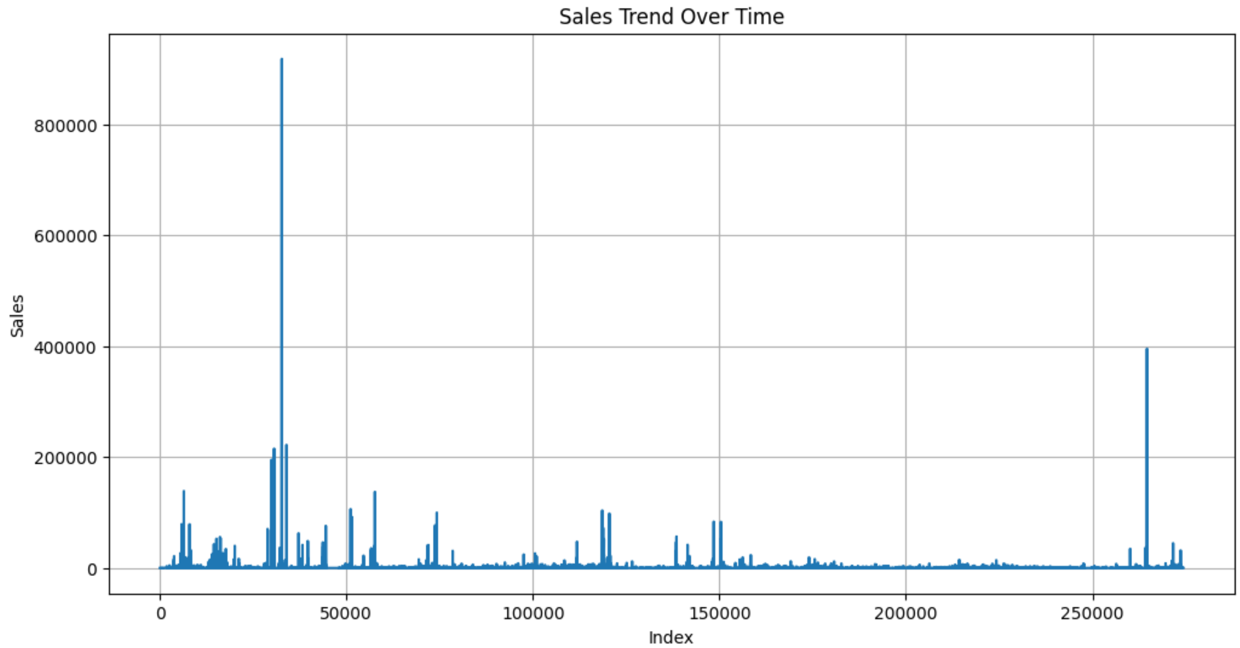

```
prediction_df = pd.DataFrame({
    'Quantity': future_quantity['sales_qty'],
    'Predicted_Sales': future_sales
})

prediction_df.to_csv('future_sales_predictions.csv', index=False)

print("\nPrediction file saved successfully!")
```

```
Prediction file saved successfully!
```

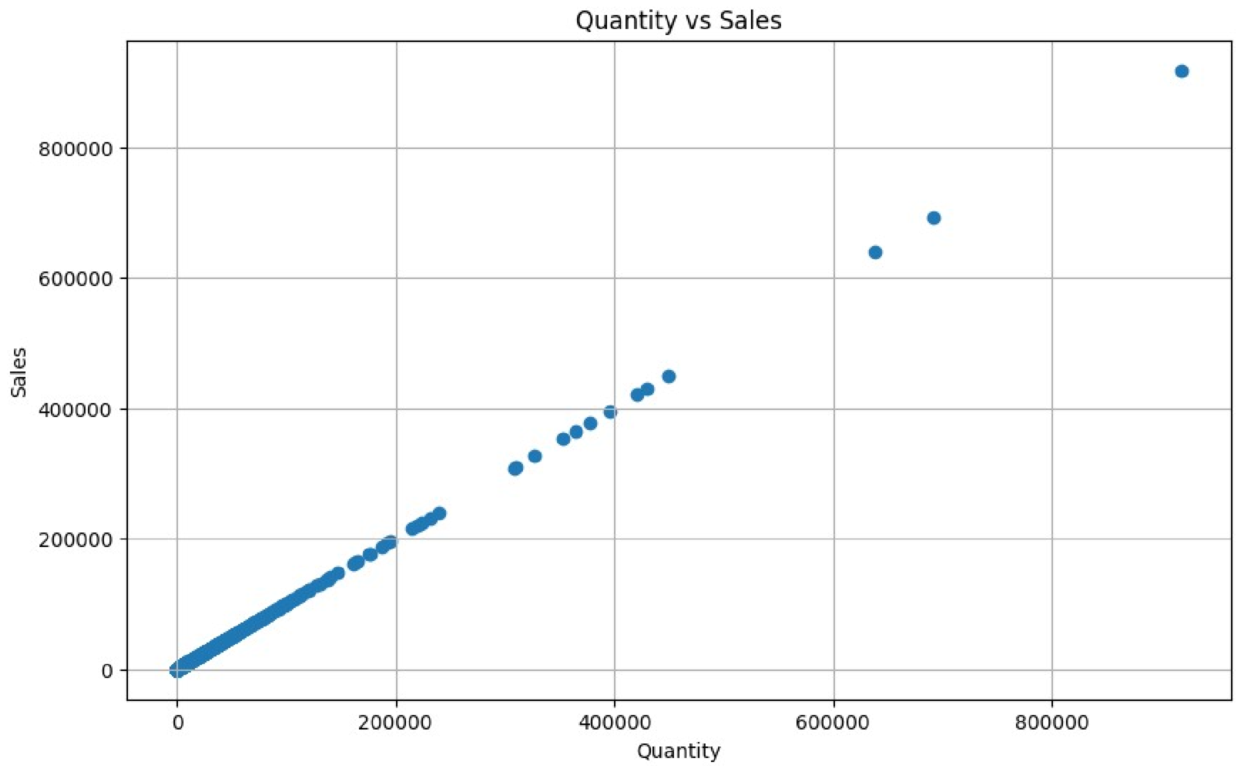
```
plt.figure(figsize=(10,6))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Sales")

plt.ylabel("Predicted Sales")

plt.title("Actual vs Predicted Sales")

plt.grid(True)

plt.show()
```



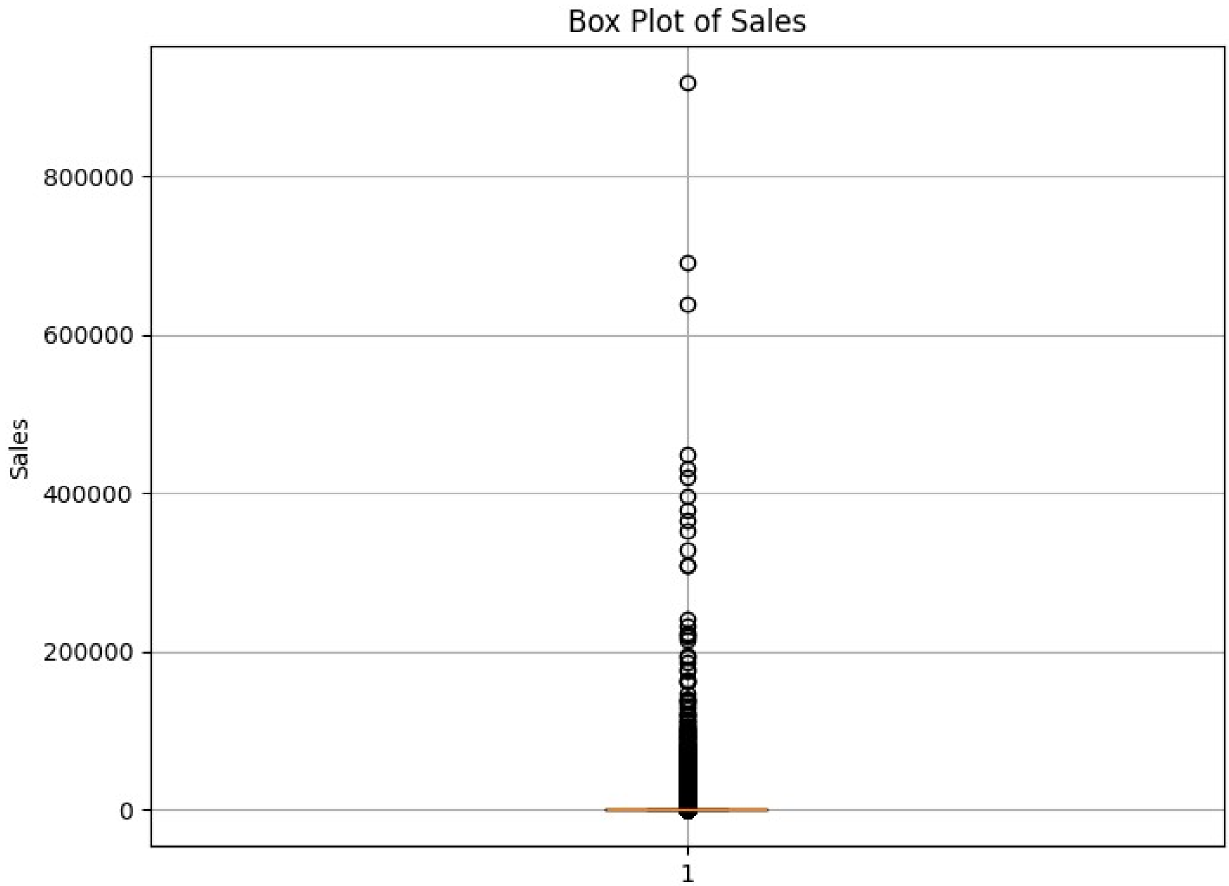

```
plt.figure(figsize=(12,6))

plt.plot(df['sales_qty'])

plt.title("Sales Trend Over Time")

plt.xlabel("Index")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



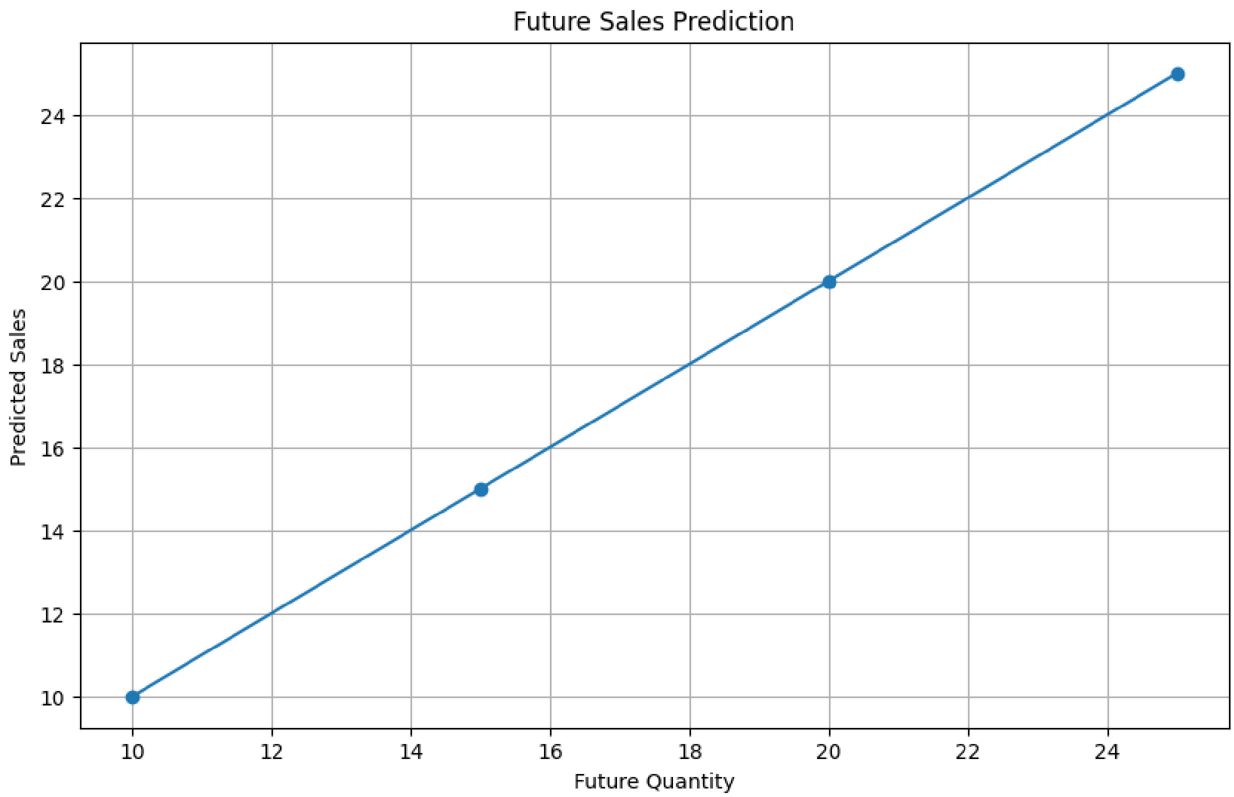

```
plt.figure(figsize=(10,6))

plt.scatter(df['sales_qty'], df['sales_qty'])

plt.title("Quantity vs Sales")

plt.xlabel("Quantity")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



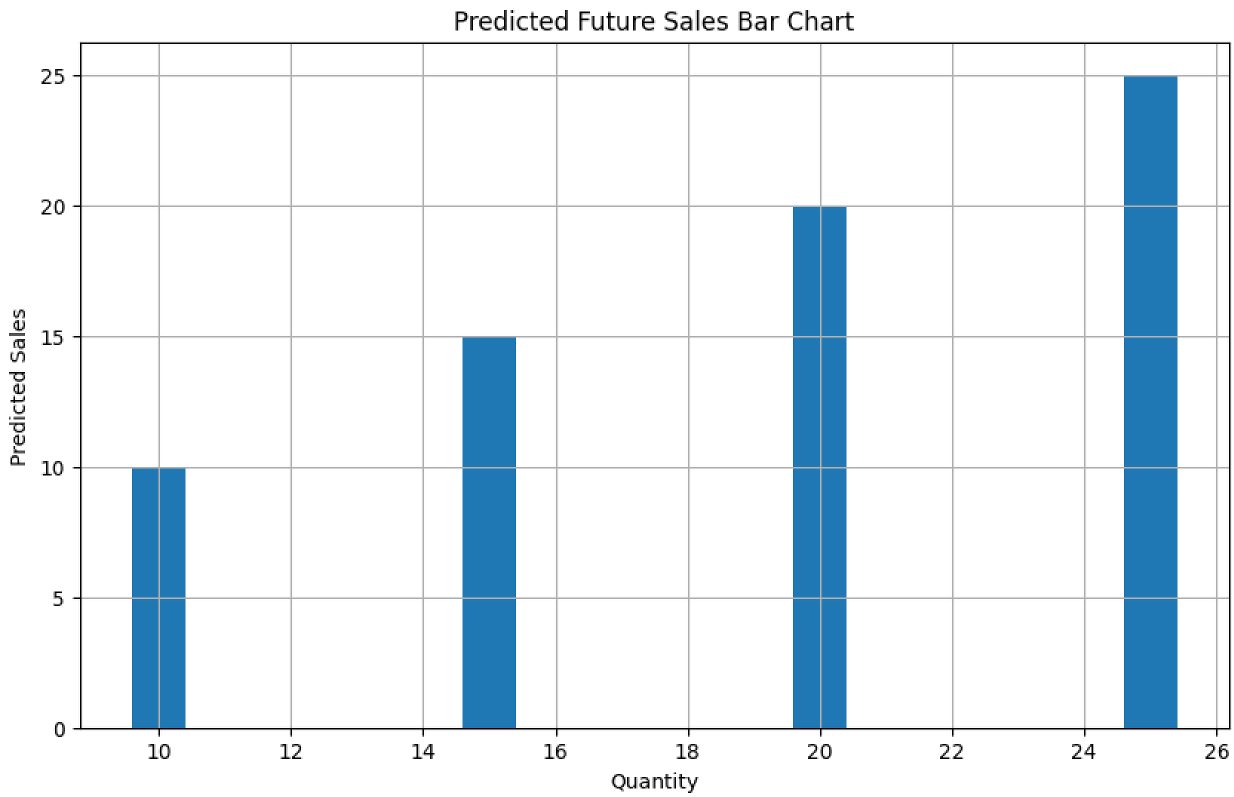

```
plt.figure(figsize=(8,6))

plt.boxplot(df['sales_qty'])

plt.title("Box Plot of Sales")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```




```
plt.figure(figsize=(10,6))

plt.plot(
    future_quantity['sales_qty'],
    future_sales,
    marker='o'
)

plt.title("Future Sales Prediction")

plt.xlabel("Future Quantity")

plt.ylabel("Predicted Sales")

plt.grid(True)
plt.show()
```




```
plt.figure(figsize=(10,6))

plt.bar(
    future_quantity['sales_qty'],
    future_sales
)

plt.title("Predicted Future Sales Bar Chart")

plt.xlabel("Quantity")

plt.ylabel("Predicted Sales")

plt.grid(True)

plt.show()
```


Predicted Future Sales Bar Chart
Predicted Sales
0
5
10
15
20
25
10
12
14
16
18
20
22
24
26
Quantity